

Layla's Shape Lab

Area & Perimeter Practice · Parallelograms · Triangles · Trapezoids · Weird Shapes

Date: _____

 Perimeter = fence AROUND → units (ft)  Area = carpet INSIDE → square units (ft²)  height = dashed $\perp$

A. Area or Perimeter? Circle A or P.

 Fence around a garden **A** **P**

 Paint a wall **A** **P**

 Ribbon around a frame **A** **P**

 Tiles on a floor **A** **P**

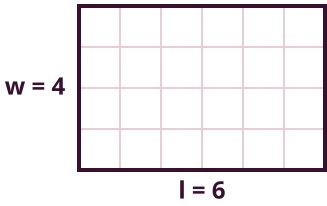
 An answer of 36 ft² **A** **P**

 An answer of 22 m **A** **P**

B. Warm-up Each square = 1 unit.

7

Count it



★ Every rectangle, always: $A = l \times w$ · $P = 2 \times (l + w)$

A = _____ units² **P** = _____ units

 Write it every time

1) formula → 2) plug in numbers → 3) answer + unit

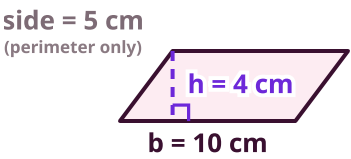
Example: $A = b \times h = 8 \times 3 = 24 \text{ cm}^2$

Slanted side? **Perimeter only.**

C. Parallelograms $A = b \times h$

8

Find A and P

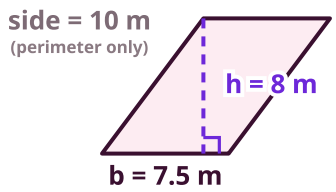


★ Every parallelogram, always: $A = b \times h$ · $P = \text{add all 4 sides}$

A = _____ **P** = _____

9

Find A

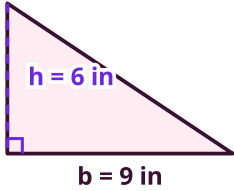


★ Every parallelogram, always: $A = b \times h = _ \times _ = _$

A = _____

D. Triangles $A = \frac{1}{2} \times b \times h$

10 **Right**



$h = 6 \text{ in}$
 $b = 9 \text{ in}$

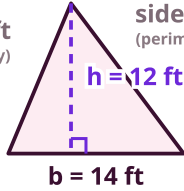
★ Every triangle, always: $A = \frac{1}{2} \times b \times h$

$A = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

11 **Find A and P**



side = 13 ft
(perimeter only)

side = 15 ft
(perimeter only)

$h = 12 \text{ ft}$

$b = 14 \text{ ft}$

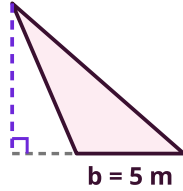
★ Every triangle, always: $A = \frac{1}{2} \times b \times h$
 $\times h \cdot P = \text{add 3 sides}$

$A = \underline{\hspace{2cm}}$ $P = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

12 **Height outside!**



$h = 7 \text{ m}$

$b = 5 \text{ m}$

★ Every triangle, always: $A = \frac{1}{2} \times b \times h$

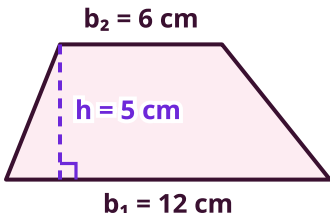
$A = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

E. Trapezoids $A = \frac{1}{2} \times (b_1 + b_2) \times h$

13 **Find A**



$b_2 = 6 \text{ cm}$

$h = 5 \text{ cm}$

$b_1 = 12 \text{ cm}$

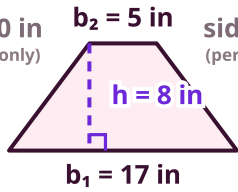
★ Every trapezoid, always: $A = \frac{1}{2} \times (b_1 + b_2) \times h$

$A = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

14 **Find A and P**



side = 10 in
(perimeter only)

$b_2 = 5 \text{ in}$

side = 10 in
(perimeter only)

$h = 8 \text{ in}$

$b_1 = 17 \text{ in}$

★ Every trapezoid, always: $A = \frac{1}{2} \times (b_1 + b_2) \times h$ · $P = \text{add all 4 sides}$

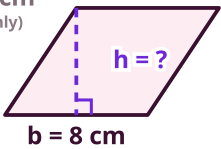
$A = \underline{\hspace{2cm}}$ $P = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

F. Backwards Given the area, find the missing part.

15 $A = 48 \text{ cm}^2$



side = 7.2 cm
(perimeter only)

$h = ?$

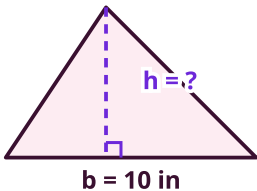
$b = 8 \text{ cm}$

★ Every parallelogram, always: $A = b \times h \rightarrow h = A \div b$

$h = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

16 $A = 30 \text{ in}^2$



$h = ?$

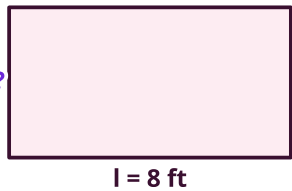
$b = 10 \text{ in}$

★ Every triangle, always: $A = \frac{1}{2} \times b \times h \rightarrow h = A \times 2 \div b$

$h = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

17 **Rectangle: $P = 26 \text{ ft}$**



$w = ?$

$l = 8 \text{ ft}$

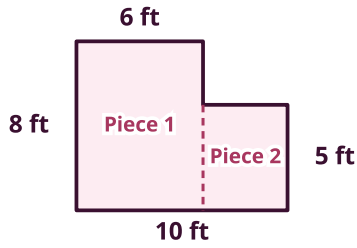
★ Every rectangle, always: $P = 2 \times (l + w)$

$w = \underline{\hspace{2cm}}$

$\underline{\hspace{2cm}}$

G. Weird shapes Split it $+$ or cut it out $-$

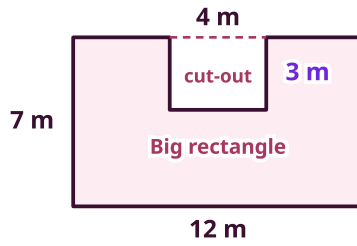
18 Find A and P



★ Weird shape, always: $A = A_1 + A_2$
P = all 6 sides

A = _____ P = _____

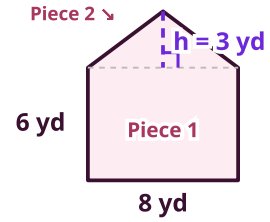
19 Big – cut-out



★ Weird shape, always: $A = \text{big} - \text{cut-out}$

A = _____

20 Rectangle + triangle



★ Weird shape, always: $A = A_1 + A_2$

A = _____

H. Word problems Draw it. Label it. Answer + unit!

21 
A garden is **12.5 ft** long and **6 ft** wide.

- a) How much fence goes around it?
b) How much soil covers it?

★ Every rectangle, always: $P = 2 \times (l + w)$ · $A = l \times w$

a) _____ b) _____

22 
A triangle pennant has a base of **10 in** and a height of **16 in**. How much felt is needed?

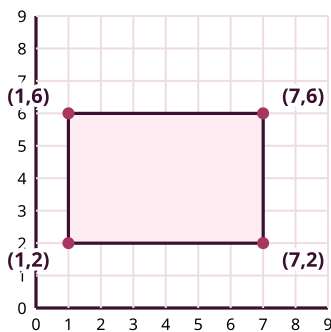
★ Every triangle, always: $A = \frac{1}{2} \times b \times h$

23 
A window is a trapezoid. Bases: **9 ft** and **5 ft**. Height: **4 ft**. How much glass?

★ Every trapezoid, always: $A = \frac{1}{2} \times (b_1 + b_2) \times h$

I. Coordinate plane

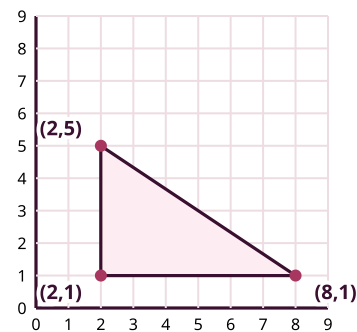
24 Rectangle



★ Every rectangle, always: $l = \text{big} - \text{small}$ · $A = l \times w$ · $P = 2 \times (l + w)$

A = _____ P = _____

25 Right triangle



★ Every triangle, always: $b, h = \text{big} - \text{small}$ · $A = \frac{1}{2} \times b \times h$

A = _____

🦋 Layla's Cheat Sheet 🦋

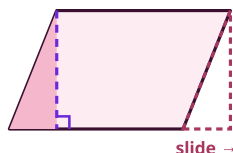
Area & Perimeter — keep this next to you!



Fence vs Carpet

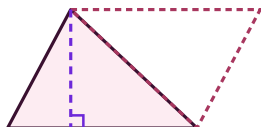
Perimeter = walk AROUND, add every outside side → **ft** (★ *always*)

Area = cover INSIDE → **ft²** (little ²)



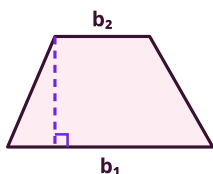
★ Every parallelogram, always: $A = b \times h$

"Slide the slice" — cut the triangle off, slide it over → a rectangle.



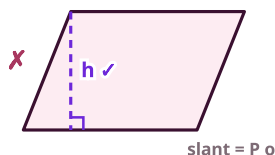
★ Every triangle, always: $A = \frac{1}{2} \times b \times h$

"A triangle is half a parallelogram." Or $b \times h \div 2$.



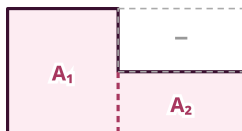
★ Every trapezoid, always: $A = \frac{1}{2} \times (b_1 + b_2) \times h$

"Add the parallels, times height, cut in half." (2 copies = 1 parallelogram)



Height stands up straight ▴

Use the dashed height with the square corner. Slanted side → **perimeter only**. Height can be **outside** an obtuse triangle.



★ Weird shape, always: split **+** or cut **-**

Split into shapes you know and add, OR big rectangle – the missing piece. Find missing sides: opposite sides match.

★ Only the numbers change. The formula for each shape never does.

⚠ **Watch out:** slanted side $\neq$ height · don't forget the $\frac{1}{2}$ · ft vs ft² · height is never part of the perimeter · count every side on L-shapes (none twice)

Backwards: $A = b \times h \rightarrow h = A \div b$. Triangle: $h = A \times 2 \div b$. **Coordinates:** length = big – small. **Always:** formula → numbers → answer + unit 🖋

Answer Key — Layla's Shape Lab

Check AFTER you try. Look at the work, not just the answer.

#	Answer	Work
A1	P	fence goes around
A2	A	covers the wall
A3	P	goes around the edge
A4	A	covers the floor
A5	A	square units → area
A6	P	plain units → perimeter
7	A = 24 units²; P = 20 units	$A = 6 \times 4 = 24$; $P = 6+4+6+4 = 20$
8	A = 40 cm²; P = 30 cm	$A = b \times h = 10 \times 4 = 40 \text{ cm}^2$ (not $\times 5$). $P = 10+5+10+5 = 30 \text{ cm}$ (not 4)
9	A = 60 m²	$A = 7.5 \times 8 = 60 \text{ m}^2$ (10 m slant not used)
10	A = 27 in²	$\frac{1}{2} \times 9 \times 6 = 54 \div 2 = 27 \text{ in}^2$
11	A = 84 ft²; P = 42 ft	$A = \frac{1}{2} \times 14 \times 12 = 84 \text{ ft}^2$ (13 & 15 not used). $P = 14+13+15 = 42 \text{ ft}$ (12 not used)
12	A = 17.5 m²	$\frac{1}{2} \times 5 \times 7 = 35 \div 2 = 17.5 \text{ m}^2$ (height is outside, still use it)
13	A = 45 cm²	$\frac{1}{2} \times (12+6) \times 5 = \frac{1}{2} \times 18 \times 5 = 45 \text{ cm}^2$
14	A = 88 in²; P = 42 in	$A = \frac{1}{2} \times (17+5) \times 8 = \frac{1}{2} \times 22 \times 8 = 88 \text{ in}^2$. $P = 17+5+10+10 = 42 \text{ in}$
15	h = 6 cm	$48 = 8 \times h \rightarrow h = 48 \div 8 = 6 \text{ cm}$
16	h = 6 in	$30 = \frac{1}{2} \times 10 \times h \rightarrow 30 = 5 \times h \rightarrow h = 6 \text{ in}$
17	w = 5 ft	$26 - 8 - 8 = 10 \rightarrow w = 10 \div 2 = 5 \text{ ft}$
18	A = 68 ft²; P = 36 ft	Missing sides: $8-5 = 3 \text{ ft}$, $10-6 = 4 \text{ ft}$. $A = 6 \times 8 + 4 \times 5 = 48 + 20 = 68 \text{ ft}^2$ (or $10 \times 8 - 4 \times 3 = 68$). $P = 6+3+4+5+10+8 = 36 \text{ ft}$
19	A = 72 m²	$12 \times 7 - 4 \times 3 = 84 - 12 = 72 \text{ m}^2$
20	A = 60 yd²	$8 \times 6 + \frac{1}{2} \times 8 \times 3 = 48 + 12 = 60 \text{ yd}^2$
21	a) 37 ft b) 75 ft²	a) $P = 12.5+6+12.5+6 = 37 \text{ ft}$; b) $A = 12.5 \times 6 = 75 \text{ ft}^2$
22	80 in²	$\frac{1}{2} \times 10 \times 16 = 80 \text{ in}^2$
23	28 ft²	$\frac{1}{2} \times (9+5) \times 4 = \frac{1}{2} \times 14 \times 4 = 28 \text{ ft}^2$
24	A = 24 units²; P = 20 units	length $7-1 = 6$, height $6-2 = 4 \rightarrow A = 24$, $P = 6+4+6+4 = 20$
25	A = 12 units²	base $8-2 = 6$, height $5-1 = 4 \rightarrow \frac{1}{2} \times 6 \times 4 = 12$